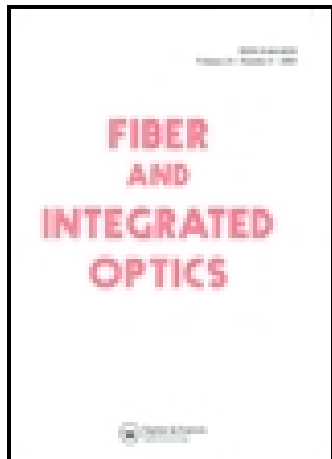




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Refractive index measurements on single-mode fiber as functions of product parameters, tensile stress, and temperature

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Refractive Index Measurements on Single-Mode Fiber as Functions of Product Parameters, Tensile Stress, and Temperature

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Abstract *This paper describes results of a recent study that characterizes effective index of refraction for commercially available matched-clad, single-mode optical fibers. Effects of product parameter, stress, and temperature ranges on the index are reported. Direct measurements of flight times, fiber lengths, tensile loads, elongations, and temperatures provide index of refraction accuracy to four decimal places. Nominal effective index of refraction values found for unshifted fiber were 1.4675 and 1.4681 at 1300 nm and 1550 nm wavelengths, respectively; values for dispersion-shifted fiber were 1.4718 and 1.4711. Both fiber types at each wavelength displayed index versus stress slopes of $-0.000024/\text{kpsi}$, and index versus temperature slopes of $0.000012/^{\circ}\text{C}$.*

Introduction

The ratio of the speed of light in a vacuum to the speed of a transmitted pulse of light in an optical fiber yields an effective index of refraction. Understanding effective index of refraction (EIOR) dependence on product parameter variations, tensile stress, and temperature has numerous applications in the fiber and fiber-related industries. For a variety of reasons, such measurements require sample fiber lengths of tens of meters or more. In order to measure the subtle influence of product variations on index, at least a four-decimal place accuracy is necessary.

Fiber Types Measured

Standard unshifted and dispersion-shifted, matched-clad single-mode optical fibers were measured at the nominal operating wavelengths of 1300 nm and 1550 nm. Samples were selected that represented a typical manufactured product range for delta (Δ) and mode-field diameter (MFD) parameters.

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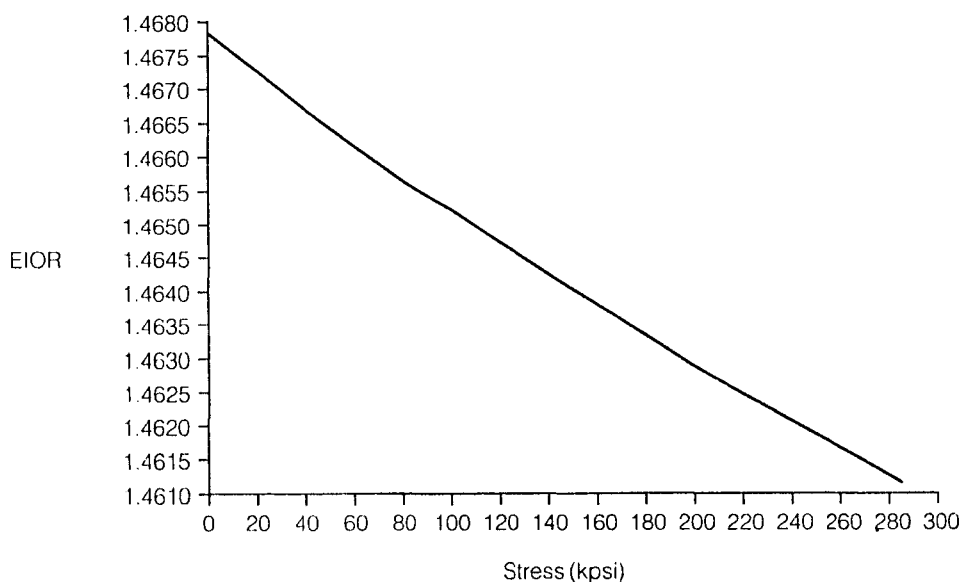


Figure 1. Effective index versus stress (unshifted single-mode—1295 nm).

Measurement Methods

Sample lengths, L_o , were deployed in a straight line and measured between 50 m benchmarks surveyed to within an accuracy of two mm. One end of the fiber was coupled to a Fresnel reflection optical time domain reflectometer (OTDR) capable of 10-picosecond time of flight resolution, $\Delta\tau$. The associated index measurement resolution, ΔN , follows using Equation (1), where c is the speed of light in vacuum (3×10^8 m/s).

$$\Delta N = (c/2L_o) \Delta\tau \quad (1)$$

Evaluating Equation (1) yields a resolution of 0.00003, assuring confidence in reporting the measured index to the fourth decimal place. The initial stage of the work involved an index of refraction characterization measurement on each sample fiber in a relaxed (unstressed) condition at room temperature (22 °C).

Once characterized, each sample was prepared in the same deployment setting for index versus stress measurements. Both fiber ends were epoxied to grooved aluminium blocks. The OTDR end was fixed, while tensile loads were applied in approximately 0.4 lb (20 kpsi) increments to the free end. Fiber elongations under load were measured to determine strain and the corrected overall length for index computations.

Refractive index as a function of temperature comprised the third stage of measurements. Characterized fiber samples were wound in loose coils and placed in a temperature chamber. Initially, index measurements were performed at room temperature to serve as controls and confirm any effect due to the coiled configuration. Measurements were taken over a temperature range of -60 to $+85$ degrees Celcius in 10° steps. Length changes, ΔL , due to thermal expansion and contraction were accounted for in the index calculations applying equation (2). The coefficient of linear thermal expansion, α , for fused silica is $8 \times 10^{-7}/^\circ\text{C}$ and the temperature change, ΔT , is measurement temperature minus room temperature.

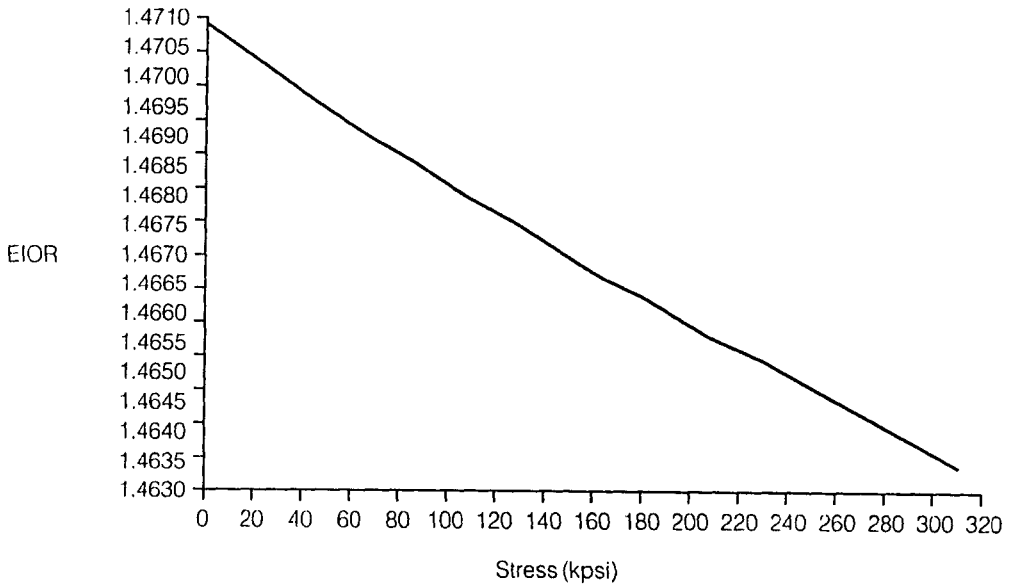


Figure 2. Effective index versus stress (dispersion shifted single-mode—1532 nm).

$$\Delta L = L_o \alpha \Delta T \quad (2)$$

Results

Characterization measurements over the entire sample range for Δ and mode-field diameter revealed index of refraction variations in the fourth decimal place for unshifted fibers and in the third decimal place for dispersion-shifted fibers. Unshifted samples exhibited an EIOR range of 0.0006 at both 1300 nm and 1550 nm wavelengths of operation, with average values of 1.4675 and 1.4681, respectively. The dispersion-

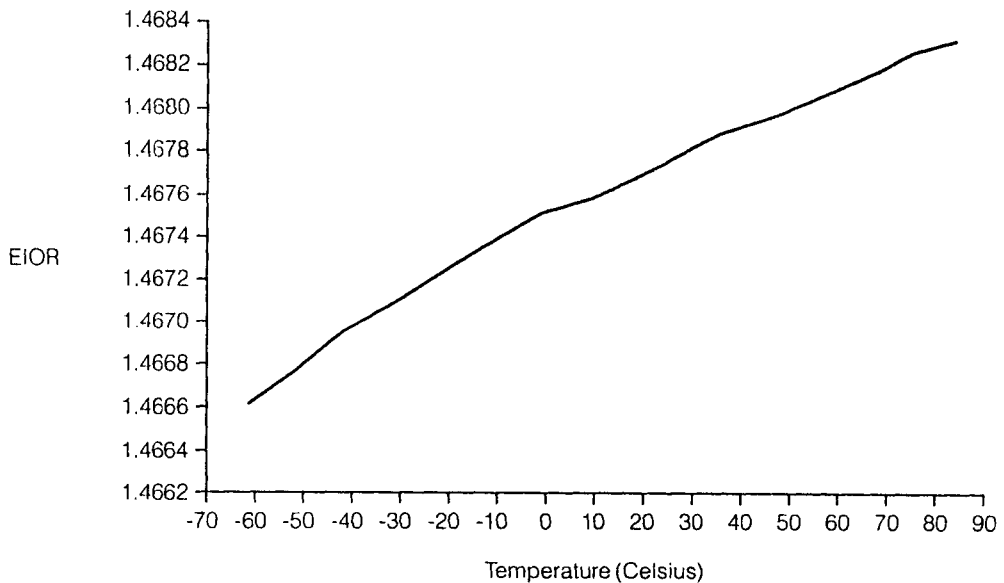


Figure 3. Effective index versus temperature (unshifted single-mode—1295 nm).

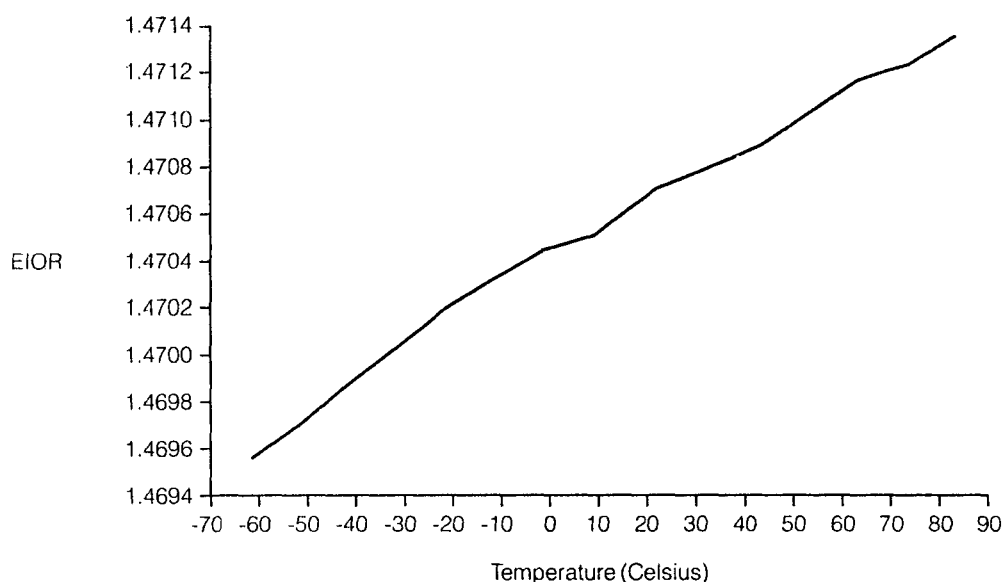


Figure 4. Effective index versus temperature (dispersion shifted single-mode—1352 nm).

shifted EIOR sample range was 0.0017 for both 1300 nm and 1550 nm wavelengths; corresponding average refractive indices were 1.4718 and 1.4711.

Example graphs, Figs. 1 and 2, of EIOR versus tensile stress for an unshifted and dispersion-shifted fiber sample show nearly linear relationships. Slopes for both fiber types were consistent at $-0.000024/\text{kpsi}$ ($-0.0027/\text{percent strain}$) across all samples measured at either wavelength. Negative slope means flight time, and therefore, index decreases with increasing tensile stress as photoelastic theory predicts [1]. The magnitude of the slope also is close to the $-0.000035/\text{kpsi}$ value one would obtain from a previous study on multimode fiber (P_2O_5/GeO_2 doped silica core), by Hartog, Conduit and Payne [2].

Plots shown in Figs. 3 and 4 are representative of EIOR versus temperature results for unshifted and dispersion-shifted fiber types. Again, all samples measured at either wavelength displayed strong consistency in the rate of change of index with temperature, $0.000012/^\circ\text{C}$. Hartog et al. also measured temperature effects on fiber index [2], obtaining data that translates into a slope of $0.000010/^\circ\text{C}$. Cohen and Fleming [3] reported $0.000010/^\circ\text{C}$ as well, measuring a depressed clad single-mode fiber (with a B_2O_3 doped silica core).

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